# An Ultra-wideband Vivaldi Antenna with Stable Radiation and Balanced Feeding Techniques



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***Abstract*—A miniaturized double-exponentially tapered Vivaldi antenna for 3-18 GHz ultra-wideband (UWB) applications is proposed in this paper, which is fed by an optimized microstrip-to-slot line transition. It achieves a 6:1 bandwidth ratio through three key innovations: (1) a compact broadband balun with a graded coplanar strip line (CPS) impedance transformer that makes VSWR less than 2 at frequencies below 3 GHz, (2) a balanced feed mechanism combining a half-wavelength extended current slot and quarter-wavelength dumbbell-shaped choke to equalize ground-to-radiator current distribution and prevent pattern distortion, (3) optimized integration of these techniques minimizing gain reduction while maintaining impedance matching. Based on a 0.813-mm thick RO4003C substrate, the integrated feed occupies only 0.21 × 0.17 $\lambda_0^2$ at 3 GHz. Simulation results reveal a VSWR of less than 2 across the 3-18 GHz range, with a peak gain of 11 dBi at 18 GHz and gain fluctuations of under 3 dB. Additionally, the antenna exhibits stable radiation patterns, thereby advancing UWB antenna technology by effectively addressing impedance matching, miniaturization, and radiation stability.**

***Keywords—Vivaldi antenna, ultra-wideband (UWB), current distribution equalization, dumbbell-shaped chokes***

## I. INTRODUCTION

Ultra-wideband (UWB) Vivaldi antennas are pivotal for radar, communications, and imaging systems due to their wide bandwidth and end-fire radiation, such as the work by Sandi et al. on microstrip array antennas for 5G [1]..Recent studies focus on enhancing gain, bandwidth, and miniaturization but exhibit inherent trade-offs. Double-slot structures boost gain to 15.5 dBi at 3 GHz but require large apertures, limiting compactness [2]. Microstrip-to-slot transitions in mm-wave designs achieve 70% bandwidth yet suffer from radiation loss and complex integration [3]. Folded-taper miniaturization improves sub-GHz matching but compromises gain uniformity [4]. Unbalanced SMA feeds used in planar designs distort radiation patterns at high frequencies [5-10].

The double-exponentially tapered Vivaldi antenna is an optimized variant of conventional Vivaldi antennas, featuring dual exponential tapers along both outer edges, making it particularly suitable for ultra-wideband applications [10]. While traditional Vivaldi antennas inherently maintain a balanced feed structure, their double-exponentially tapered counterparts typically employ unbalanced connectors and transmission lines. This configuration creates a fundamental impedance mismatch because the antenna requires balanced input currents. Consequently, an unbalanced-to-balanced converter (balun) becomes essential for proper operation. However, this solution introduces two significant challenges: (1) the balun's design complexity increases substantially with the antenna's operational bandwidth [12], and (2) the additional balun inevitably occupies valuable physical space.

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To solve the mentioned problems, this paper presents a miniaturized Vivaldi antenna operating from 3 to 18 GHz, achieving a 6:1 bandwidth ratio through three key innovations:

(1) a compact CPS-based balun enabling VSWR < 2 below 3 GHz,

(2) a balanced feeding structure incorporating a $\lambda/2$ extended slot and a $\lambda/4$ dumbbell choke for current equalization,

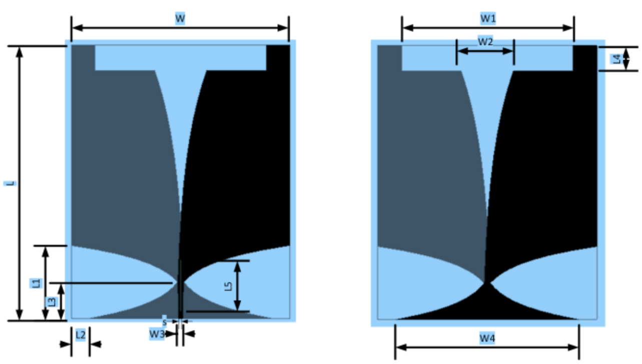
(3) an integrated design that minimizes gain degradation while maintaining radiation pattern stability.

The antenna, fabricated on a 0.813-mm RO4003C substrate, occupies only $0.21\times0.17\ \lambda_0^2$ at 3 GHz. Simulation results confirm a VSWR < 2 across the band, a peak gain of 11 dBi at 18 GHz, and gain variation below 3 dB. Although currently validated by simulation, the design is suitable for standard PCB fabrication and will be prototyped in future work for experimental verification.

## II. Configuration and Design of the Proposed Antenna

### A. The Structure of the Proposed Antenna

The 2D structure of the proposed antenna is shown in Fig. 1. The proposed antenna is printed on a 0.813-mm RO4003C substrate. The top layer is depicted in Fig. 1(a), where a single exponentially tapered copper element, marked in black, serves as one of the radiation sources for the antenna. The bottom layer is shown in Fig. 1(b), another exponentially tapered copper is marked in gray. The graded ground plane structure is a key factor in achieving wideband performance [11]. The antenna's optimized parameters are listed in Table I.




(a) (b)

Fig. 1. The structure of the proposed antenna, (a) top layer, (b) bottom layer.

TABLE I. The Optimized Parameters of the Proposed Antenna

| Parameters | Value | Parameters | Value |
|---|---|---|---|
| L | 75 mm | W | 60 mm |
| $L_1$ | 20 mm | $W_1$ | 47 mm |
| $L_2$ | 5 mm | $W_2$ | 14.3 mm |
| $L_3$ | 10 mm | $W_3$ | 1.15 mm |
| $L_4$ | 7 mm | $W_4$ | 50 mm |
| $L_5$ | 14 mm | S | 0.1 mm |
| T | 0.813mm | | |

### B. Feed Structure Design

The antenna employs a stepped-impedance microstrip-to-slotline transition to achieve broadband matching. The characteristic impedance $Z_{cps}$ of the coplanar stripline (CPS) section is given by [13]:

$$Z_{cps} = \frac{30\pi}{\sqrt{\varepsilon_{eff}}}\frac{K(k')}{K(k)} \tag{1}$$

where $\varepsilon_{eff}$ is the effective dielectric constant, $K$ is the complete elliptic integral, $k'$ is the complementary modulus of $k, k = W/(W+2S)$, and S is the slot width (0.1 mm). The optimized impedance transformer (Fig. 1) provides VSWR < 2 below 3 GHz through three cascaded sections with impedance ratios:

$$\Gamma_m = 20log\left|\frac{Z_m - Z_{m-1}}{Z_m + Z_{m-1}}\right| < -15dB \tag{2}$$

achieving a 6:1 bandwidth (3–18 GHz) on a 0.813-mm RO4003C substrate ($\varepsilon_r = 3.55$).

### C. Current Equalization Mechanism

To mitigate ground current imbalance caused by SMA feeding, a $\lambda/4$ dumbbell choke (at 12GHz) creates a high impedance ($Z_{choke} = 100\sim120\Omega$):

$$Z_{choke} = Z_0\sqrt{\frac{\varepsilon_{eff}+1}{2}}\cdot ln\left(\frac{S}{W_3}\right) \tag{3}$$

where $W_3 = 1.15mm(Table\ I)$. This suppresses signal-line current while enhancing ground coupling . Concurrently, the $\lambda/2$ extended slot ($L_4 = 7\ mm$) redistributes the surface current density $J_s$.

### D. Integrated Design Optimization

The synergistic integration of the feed transition, $\lambda/2$ slot, and $\lambda/4$ choke minimizes the gain degradation. Parametric optimization yields:

- Choke dimensions: $L_5 = 14$ mm , $W_3 = 1.15$ mm
- Slot dimensions: $L_4 = 7$ mm , $W_1 = 47$ mm

## III. The Working Mechanism of the Proposed Antenna

### A. The Working Mechanism of the Tapered Microstrip Feed Structure

Fig. 2 illustrates the simulated surface current distributions at three strategically selected frequencies: 3 GHz, 12 GHz, and 18 GHz. These frequencies were chosen to comprehensively evaluate the antenna's performance across its entire ultra-wide operational bandwidth (3-18 GHz, bandwidth ratio 6:1)

- 3 GHz: Represent the lower operating band limit. This is critical for validating the effectiveness of the low-frequency impedance matching innovations, particularly the half-wavelength extended slot ($L_4 = 7$ mm). The performance at the band edge often poses the greatest challenge for UWB antennas.
- 12 GHz: Serve as a central reference frequency within the operational band. Crucially, this is the design frequency (λ/4) for the dumbbell-shaped choke structure discussed later in sub-section B. Observing behavior here is essential to understand the core current balancing mechanism.

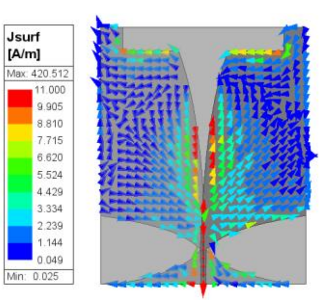
- 18 GHz: Represent the upper operating band limit. This high frequency result shows the antenna's structural integrity, potential for pattern distortion, and the stability of the feed and radiation mechanisms under more demanding conditions.

The tapered microstrip feed structure, coupled with the gradient-changed ground plane, is fundamental in achieving broadband impedance matching and facilitating the transition from the microstrip feed line to the radiating slot line. This design ensures efficient energy transfer across the vast frequency range.

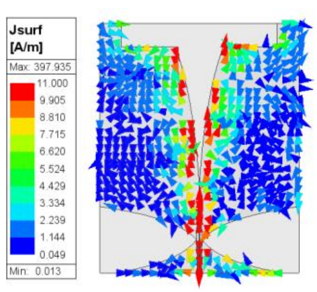
A key innovation working synergistically with this feed structure is the half-wavelength extended slot ($\mathrm{L}_4 = 7$ mm). Its impact is most pronounced at the lower frequencies, as particularly evident in the 3 GHz current distribution shown in Fig. 2(a). At this frequency, a significant portion of the current is confined within the region defined by this extended slot. While the primary radiating currents still concentrate along the characteristic edges of the Vivaldi structure, this confinement within the slot region is a direct consequence of its resonant behavior. The half-wavelength dimension ($L_4$) at low frequencies creates a resonant structure that effectively traps and manipulates currents, thereby dramatically improving the impedance matching performance of the lower end of the UWB spectrum.

Across all three frequencies (3, 12, and 18 GHz), the current distributions exhibit a highly desirable and consistent pattern: currents are predominantly concentrated along the exponentially tapered edges of the Vivaldi antenna. The distribution consistently shows a characteristic half-wavelength standing wave pattern, with maximum current intensity occurring at the center of the radiating edges and gradually attenuating towards the ends. This uniformity across the 6:1 bandwidth is a clear indicator of the successful excitation of the fundamental half-wavelength directional radiation mode.

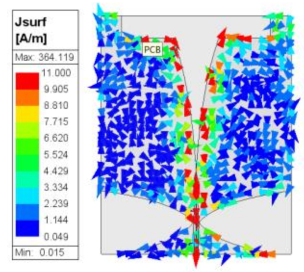
The stability and uniformity of this current behavior from the low band edge (3 GHz) through the design center (12 GHz) to the high band edge (18 GHz) provide compelling evidence for the robustness of the integrated feed design (tapered microstrip, gradient ground, and half-wavelength slot). It confirms the antenna's ability to maintain stable radiation characteristics and achieve true broadband operation throughout the entire 3-18 GHz range. The consistent excitation of the desired mode underpins the observed stable gain and radiation patterns presented later in Section IV.




(a)




(b)

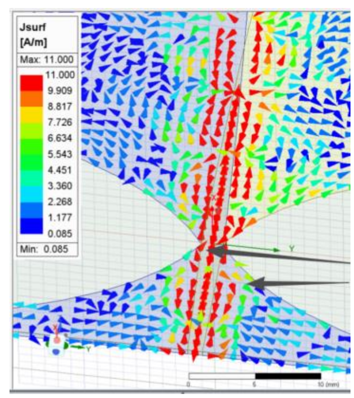



(c)

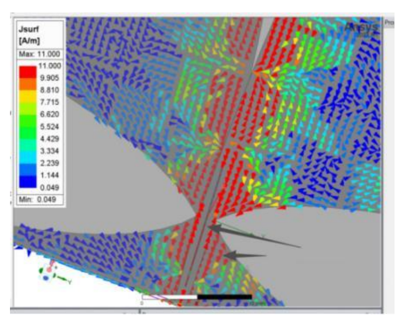
Fig. 2. Simulated surface current distributions at (a) 3 GHz, (b) 12 GHz, and (c) 18 GHz, demonstrating consistent excitation of the fundamental half-wavelength mode across the 6:1 bandwidth.

## B. *The Working Mechanism of the Current Equalization Techniques*

To address the inherent current imbalance between signal line and ground plane caused by SMA feed excitation, this study proposes an innovative quarter-wavelength (λ/4 at 12 GHz) dumbbell-shaped choke structure. Fig. 3 presents a comparative analysis of simulated current distributions with and without this balancing mechanism. In the baseline configuration (without choke and half-wavelength slot), the current distribution shows significant asymmetry—most of the total current is concentrated on the top radiating arm, while the bottom carries only a small amount of current.




(a)



(b)

Fig. 3. The simulated current distributions at 12GHz, (a) traditional Vivaldi antenna, (b) the proposed antenna.

The implemented choke structure introduces a deliberate impedance discontinuity (Z ≈ 100-120 Ω at 12 GHz) at a strategic location. This engineered mismatch produces two beneficial effects: (1) it effectively suppresses the traveling-wave current amplitude on the signal line, and (2) simultaneously enhances ground plane current coupling by through controlled electromagnetic field redistribution. Significantly optimizing the conditions at the SMA feed point.

## IV. Results and Discussion

Fig. 4 displays the simulated VSWR and gain performance of the proposed antenna. The gradient-changed ground enables an effective microstrip-to-slotline transition, achieving VSWR < 2 across the entire 3-18 GHz bandwidth. The antenna attains a peak gain of 11 dBi at 18 GHz, with gain fluctuations maintained below 3 dB in the higher frequency band.

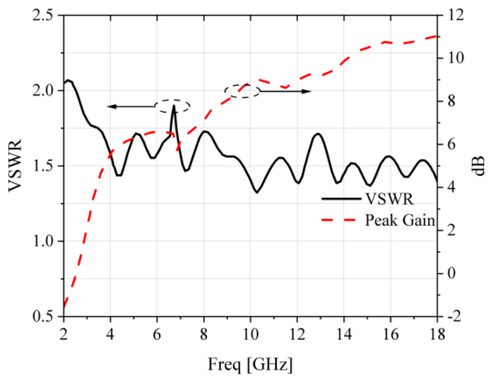
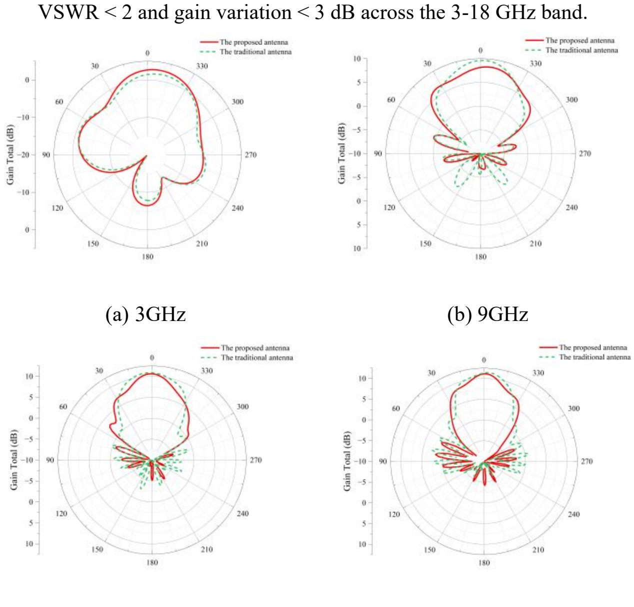



Fig. 4. Simulated VSWR and peak gain of the proposed antenna, confirming a VSWR < 2 and gain variation < 3 dB across the 3-18 GHz band.



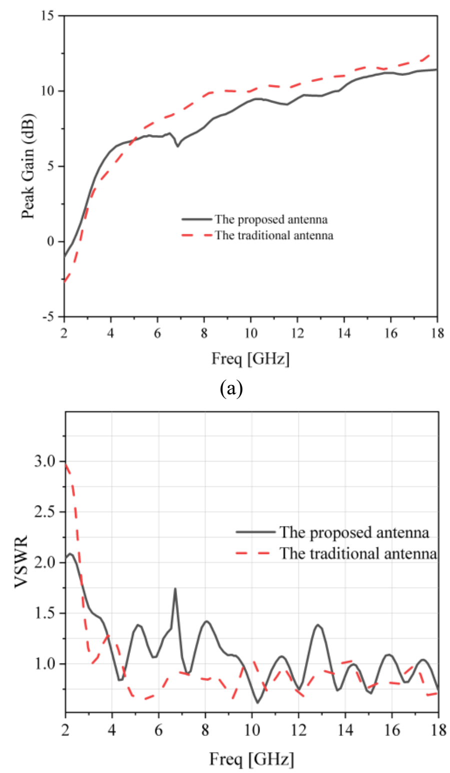

(a) 3GHz (b) 9GHz

(c) 12GHz (d )18GHz

Fig. 5. Comparison of the H-plane radiation patterns between traditional Vivaldi antennas and the proposed antenna at (a) 3GHz, (b) 9GHz, (c) 15GHz, (d) 18GHz.

Fig. 5 shows a comparison of the radiation patterns in the H-plane between the traditional Vivaldi antenna and the proposed antenna. It can be observed that after adding the choke and the half-wavelength slot, the antenna maintains consistent directional radiation across the entire operating frequency band, exhibiting lower backlobe levels and well-controlled sidelobes, successfully reducing diagram distortion. Crucially, there is no observable degradation in the radiation pattern throughout the operational frequency range, confirming stable beamforming capability.

Fig. 6 shows trade-off between improved balance and radiation efficiency. The modified design shows a controlled gain reduction of 1.2-1.8 dB across the operational bandwidth. Through parametric optimization of the choke's geometric parameters (particularly slot width = 0.8 mm and length = 6.25 mm at 12 GHz), this performance degradation is kept within acceptable limits (<15% reduction in efficiency) while ensuring stable radiation patterns. The final design achieves an excellent compromise between feed balance (VSWR < 2) and radiation performance (gain variation < 3dB) throughout the 3-18 GHz range.




(a)

(b)

Fig. 6. The comparison of traditional Vivaldi antenna and the proposed antenna, (a) gain, (b) VSWR.

Table II presents a performance comparison of ultra-wideband vivaldi antenna reported in recent years. The proposed antenna demonstrates significant performance advantages in ultra-wideband Vivaldi antenna design, particularly in achieving an optimal balance between bandwidth, compactness, and electrical performance. While all compared works achieve a VSWR < 2, this design offers the widest absolute operational bandwidth (3-18 GHz, 142.8% FBW), covering a significantly larger frequency span (15 GHz) than most references. Crucially, it maintains this exceptional bandwidth while achieving a highly compact size (75 mm × 60 mm, 0.21×0.17 $\lambda_0^2$ at 3 GHz). This miniaturization is superior to most wideband designs.

Furthermore, the antenna delivers robust gain performance (11 dBi peak gain at 18 GHz) with minimal variation (<3 dB) across the entire band. While Ref. [2] reports a higher peak gain (15.5 dBi), it achieves this over a much narrower bandwidth (0.3-3 GHz) and with a vastly larger aperture. The combination of the widest absolute bandwidth, competitive peak gain, excellent gain stability, and a highly compact footprint represents a significant advancement in UWB Vivaldi antenna technology, effectively addressing the inherent trade-offs between bandwidth, size, and gain uniformity prevalent in prior works.

TABLE II. PERFORMANCE SUMMARY AND COMPARISON WITH OTHER WORKS

| | **this work** | **Ref. [4]** | **Ref. [2]** | **Ref. [10]** | **Ref. [8]** | **Ref. [5]** | **Ref. [6]** | **Ref. [7]** | **Ref. [3]** |
|---|---|---|---|---|---|---|---|---|---|
| VSWR | **<2** | <2 | <2 | <2 | <2 | <2 | <2 | <2 | <2 |
| BW (GHz) | **3-18** | 0.3-6 | 0.3-3 | 1.28-12 | 1-11.4 | 2-32 | 3-11 | 0.7-7.3 | 24.17-55 |
| FBW(%) | **142.8** | 180 | 164 | 159.97 | 168 | 175 | 114 | 166 | 78.5 |
| Peak gain(dBi) | **11** | 9.2 | 15.5 | 4 | 10.18 | 12.1 | 9.2 | 11.2 | 5 |
| Size ($mm^2$) | **75×60** | 213.5×305 | 460×400 | 53×34 | 224×300 | 140×66 | 90×63 | 240×220 | 3.9×3.9 |

## V. CONCLUSION

This study introduces a miniaturized Vivaldi antenna for 3-18 GHz UWB applications. Key innovations include: (1) A broadband balun with CPS transformer (VSWR<2 below 3GHz), (2) Balanced feed using λ/2 slot and λ/4 choke for current equalization, and (3) Optimized integration for stable performance. Based on a 0.813-mm-thick RO4003C substrate, the antenna achieves a 6:1 bandwidth with VSWR<2, a peak gain of 11 dBi at 18 GHz, gain variation below 3 dB, and stable radiation patterns (front-to-back ratio >8 dB). The compact design (0.21×0.17$\lambda_0^2$ at 3 GHz) simultaneously addresses the challenges of impedance matching, miniaturization, and radiation stability.

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